

Problem 3.1

$$AB = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix} \quad AC = \begin{bmatrix} 10 & 12 \\ 20 & 24 \end{bmatrix}$$

$$AB + AC = \begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$$

$$B + C = \begin{bmatrix} 1 & 0 \\ 5 & 6 \end{bmatrix}$$

$$A(B + C) = \begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$$

$$AB + AC = A(B + C)$$

Problem 3.2

$$\left[\begin{array}{ccc|ccc} 1 & a & b & 1 & 0 & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & a & b & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & a & 0 & 1 & 0 & -b \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -a & -b + ac \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$